MCTR I
Lateral Movement

Work Package Scope of Work — Engineering Design 2026

CONSTRAINT: DC motor cost < R250

Main Work Elements

1. **Mechanism design —** trolley/carriage on the rail, wheel and bearing mounting, custom 3D-printed wheels (feeding the wheel wear-characterization exercise), other motion components (gears, pulleys, belts) printed or bought as practical, and the overall system mounting plate as a direct output of this design work.
2. **Motor and driver selection —** COTS DC motor under the R250 constraint with a matching driver, including selection of its own COTS low-voltage supply.
3. **Drivetrain dynamics and sizing —** acceleration/deceleration modelling to justify the motor and driver choice, including the centre-of-mass and weight-distribution analysis that also drives the mounting plate layout.
4. **Controller interface —** wiring direction, PWM, and quadrature encoder signals into the IO board header already defined; confirming encoder resolution is adequate for the dashboard's motion profiles.
5. **Characterization and test —** measuring actual acceleration/deceleration against the modelled prediction, confirming cost-constraint compliance, correlating wheel wear against print parameters, and handing off the validated performance envelope to REII.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).